FIN207

Examination - July Semester 2025

Game Theory and Design Thinking

Friday, 21 November 2025 10:00 am – 12:00 pm

Time allowed: 2 hours

INSTRUCTIONS TO STUDENTS:

1. This examination contains **4** questions and comprises **5** printed pages (including cover page).
2. You must answer **4** questions.
3. All answers must be written in the answer book.
4. This is an **Open-book** examination.

At the end of the examination.
Please ensure that you have written your examination number on each answer book used.

Failure to do so will mean that your work cannot be identified.
If you have used more than one answer book, please tie them together with the string provided.

THE UNIVERSITY RESERVES THE RIGHT NOT TO MARK YOUR SCRIPT IF YOU FAIL TO FOLLOW THESE INSTRUCTIONS.

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(Full marks: 100)

Section A (100 marks)
Answer all questions in this section.

Question 1

Assume there are two AI companies operated by two competitive owners (i.e., Player A and Player B) in the tech industry. They both offer an AI-powered daily personal assistant service. The advertised price for each subscription is fixed by the regulatory authorities. However, each owner could secretly offer a discount to attract more customers.

If one provides the discount while the other does not provide the discount, the one who provides the discount will attract most of the customers and gain a profit of $8k per day while the other could only get $1k per day.

If they both choose not to provide any discount, they can evenly get the customers and earn $5k per day.

If both offer a discount, their costs increase, and they each earn $3k per day.

Consider a game between Player A and Player B where they can choose to either offer the discount (Strategy: Discount) or not (Strategy: No Discount). The payoff is the profit they earn.

Question 1a

Interpret the game as a simultaneous game by showing a game table.

(10 marks)

Question 1b

Indicate which of the following games is the most similar to the game in (a) and explain your choice.

1. Battle of sexes
2. Chicken game
3. Prisoners' dilemma
4. None of the above choices.

(10 marks)

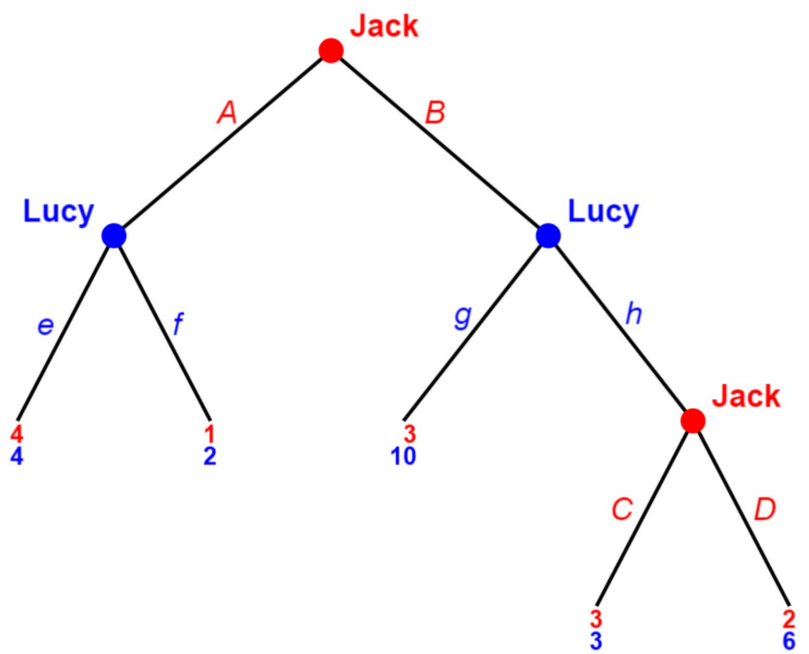
Question 1c

As one of the owners, you hope to cooperate with the other owner so that both of you choose the "No discount" strategy, allowing you to earn $5K per day. Outline ***one (1)*** way you can do to achieve this co-operation (with proper assumptions if needed) and explain how it works. **(Note: No calculation is required. Word limit: 120 words)**

(10 marks)

Question 2

Consider a sequential game between Jack and Lucy. The first payoff component belongs to Jack and the second component belongs to Lucy (e.g., if Jack chooses A and Lucy chooses f, Jack's payoff is 1 and Lucy's payoff is 2.).



Question 2a

Show each player's available strategies.

(10 marks)

Question 2b

Find and show the Rollback equilibrium(s) or explain why if there is none.

(10 marks)

Question 2c

Lucy would like to post a threat to Jack "If you select Choice A in the first move, I will select Choice F to ensure you get the worst outcome". Discuss whether this threat can help Lucy get the highest payoff. **(Word limit: 100 words)**

(10 marks)

Question 3a

A newlywed couple, Bill and Jane, recently got a 4-room HDB flat through the Built-To-Order (BTO) scheme. They are considering hiring their best friend, Jimmy, to renovate their kitchen. However, Jimmy, who recently obtained his license and launched his interior design business this year, lacks experience in such projects. Despite Jimmy's rough estimate of the renovation cost at $20,000, future uncertainty might change this figure. Jimmy talked to Bill about his estimation. Bill proposes 3 contracts below:

Contract 1: Jimmy will complete the renovation for a payment of $20,000. Should the actual cost exceed this amount during the renovation, Bill will bear the additional expenses, but no extra payment will be made to Jimmy in that scenario.

Contract 2: Jimmy completes the renovation for the estimated amount plus a buffer, which is $20,000+$5,000. Any cost overruns are Jimmy's responsibility.

Contract 3: Jimmy completes the renovation for a fixed payment of $5,000; any cost is Bill's responsibility.

Indicate which contract is closest to a fixed price contract. Explain your answer.

(10 marks)

Question 3b

Evaluate if the following 2 statements are True, False, or partially True. Explain your answer clearly.

"Incentive constraint in a mechanism design ensures that each participant's best strategy is to act truthfully or follow the intended rules of the mechanism. It also reflects that the participants should be willing to join the game."

(10 marks)

Question 4

Consider a simultaneous game between Bill and Kelly with the following game table.

		Kelly		
		D	E	F
Bill	A	10,8	0,9	5,10
	B	3,0	3,6	8,7
	C	12,3	5,5	7,4

Question 4a

Indicate whether the game table is dominance solvable and explain why.

(10 marks)

Question 4b

Assume that Bill and Kelly would like to use two strategies to form a mixed strategy Nash equilibrium. Bill will use a mixture of strategies B and C, while Kelly will use a mixture of strategies E and F. Calculate the mixed strategy Nash equilibrium.

(10 marks)

----- END OF PAPER -----